



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW5 - AUTOMATION TESTING

Software Testing Project Report

Authors:

Cao Uyển Nhi (22127310)

ltthuy22@clc.fitus.edu.vn

Supervisors:

Teacher Trần Duy Hoàng

Teacher Hồ Tuấn Thanh

Teacher Trương Phước Lộc

August 1, 2025

Table of Contents

1	Group Information	1
2	Automation Process Step by Step	1
2.1	Checkout Feature	1
2.2	Signup Feature	2
3	Implementation of Data-Driven Techniques	3
4	Use of Checkpoints/Assertions	3
5	Results of the Automated Tests	3
6	Video demonstration	4
6.1	Feature 1: User Registration	4
6.2	Feature 2: Checkout	4
7	Self-Assessment	4

1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyên Nhi	22127310	- SignUp	Done
		- Checkout	Done
Lưu Thanh Thuý	22127410	- SignIn	Done
		- User Management	Done
Nguyễn Phước Minh Trí	22127424	- Catalog	Done
		- Categories	Done
Võ Lê Việt Tú	22127435	- MyProfile	Done
		- Order Management	Done
Trần Thị Cát Tường	22127444	- Contact	Done
		- Category Management	Done

2 Automation Process Step by Step

2.1 Checkout Feature

Load Test Data

- The script reads `checkout_test_data.csv` using Pandas. Each row contains all necessary fields for a checkout scenario: product ID, quantity, billing address, payment method, bank details, and expected result.
- Data validation is performed to skip malformed rows and handle missing values with sensible defaults.

Initialize Browser

- For each supported browser (Firefox, Chrome, Safari), a new WebDriver instance is created. This ensures cross-browser compatibility.
- Log files are created per browser to record results.

Product Navigation

- The script constructs the product detail URL using the product ID from the test data and navigates directly to that page.
- Waits for Angular to stabilize before proceeding.

Add to Cart

- Simulates clicking the “Add to Cart” button using its unique selector.
- Waits for the UI to update and confirms the product is added.

Go to Cart

- Navigates to the cart page by clicking the cart icon/button.
- Waits for the cart page to load and verifies the product is present.

Update Quantity

- Updates the product quantity in the cart using the value from the test data.

- Handles potential stale element exceptions by retrying the operation.

Proceed to Checkout

- Clicks the checkout button and waits for the next page.
- If redirected to login, performs login using credentials from the script or test data.

Fill Billing Address

- Fills in all billing address fields (street, city, state, country, postcode) using test data.
- Proceeds to the next step after successful entry.

Select Payment Method

- Selects the payment method from a dropdown, matching the value from the test data (case-insensitive).
- If “Bank Transfer” is selected, fills in bank name, account name, and account number.
- For other payment methods, checks for the presence of required input fields.

Confirm Checkout

- Clicks the final confirmation button.
- Waits for a confirmation message or alert to appear, indicating the result of the checkout.

Result Logging

- Each step logs its status (PASS/FAIL) with details to both the terminal and a log file.
- Errors are captured and reported with full context for debugging.

2.2 Signup Feature

Load Test Data

- Reads `registration_test_data.csv` using Pandas. Each row contains user details (name, email, password, etc.) and expected result.
- Handles missing or malformed data gracefully.

Initialize Browser

- Creates a WebDriver instance for each browser.
- Sets up logging for each run.

Navigate to Signup Page

- Directly opens the registration page URL.
- Waits for the page and form elements to be ready.

Fill Registration Form

- Enters user details into the form fields using selectors mapped to the test data columns.
- Handles optional fields and default values.

Submit Registration

- Clicks the submit button.
- Waits for a success message, confirmation, or error alert.

Result Logging

- Logs the outcome (success or failure) for each test case, including error details and actual vs expected results.
- Results are written to both the terminal and a log file for traceability.

3 Implementation of Data-Driven Techniques

- **CSV-Driven:** Both checkout and signup scripts use Pandas to read test data from CSV files, allowing easy scaling and maintenance.
- **Dynamic Injection:** All test parameters are injected into the automation flow for each test iteration, supporting a wide range of scenarios.
- **Error Handling:** The scripts skip or report malformed data, ensuring robustness.
- **Separation of Test Logic and Data:** Test logic is generic, while data is externalized, making it easy to add, remove, or modify test cases.

4 Use of Checkpoints/Assertions

- **Element Presence/Visibility:** Before interacting with any element, the script waits for its presence or visibility, reducing flakiness.
- **Step-by-Step Validation:** After each major action (add to cart, login, fill address, select payment, submit form), the script checks for expected UI changes or messages.
- **Result Verification:** After checkout or signup, the script verifies confirmation messages, alerts, or error messages against the expected result from the test data.
- **Error Logging:** Any failure is logged with full context, including the test case ID, input data, expected result, actual result, and error details.

5 Results of the Automated Tests

Comprehensive Logging

- For each test case, the script logs every step (e.g., navigation, form fill, button click, assertion) with timestamps and browser info.
- Log files are created per browser and per feature (e.g., `logs/test_results_firefox.txt`, `logs/test_results_chrome.txt`).
- Each log entry includes: test case ID, input data, expected result, actual result, and error details if any.

PASS/FAIL Tracking

- The script compares the actual result (confirmation message, error alert, etc.) with the expected result from the test data.

- If they match, the test case is marked as PASSED; otherwise, as FAILED.
- For failed cases, the log includes the error type and the step where the failure occurred.

Traceability

- Each test case is uniquely identified by its ID and input data, making it easy to trace and reproduce issues.
- The logs provide a full history of actions and outcomes for every test run.

Cross-Browser Coverage

- By running the same test cases on multiple browsers, the report highlights any browser-specific issues or inconsistencies.
- Results are grouped by browser for easy comparison.

Summary and Analysis

- After all tests, log files can be parsed to generate summary statistics:
 - Total test cases executed
 - Number of PASSED/FAILED cases per browser and feature
 - Most common error types
 - Coverage of different scenarios (e.g., payment methods, address formats)
- These summaries help stakeholders quickly assess the quality and reliability of the checkout and signup features.

6 Video demonstration

6.1 Feature 1: User Registration

[Automation process of User Registration](#)

6.2 Feature 2: Checkout

[Automation process of Checkout](#)

7 Self-Assessment

No.	Criteria	Grade	Self-Assessed
1	Sign In	50	50
1.1	Report + Bug	15	15
1.2	Test cases	5	5
1.3	Script files	10	10
1.4	Data files	10	10
1.5	Videos	10	10
2	User Management	50	50
2.1	Report + Bug	15	15
2.2	Test cases	5	5
2.3	Script files	10	10
2.4	Data files	10	10

No.	Criteria	Grade	Self-Assessed
2.5	Videos	10	10
	Total	100	100